

MUHAMMAD HISYAM

Full Stack & AI Software Engineer

Email: muhammad2504hisyam@gmail.com | GitHub: [caleheinz25](https://github.com/caleheinz25) | LinkedIn | mhisyam.com | Bekasi, Indonesia

Full Stack & AI Software Engineer and Universitas Gunadarma Computer Science graduate. Creator of **26+ open-source projects** spanning desktop cross-platform applications, high-performance web systems, medical deep learning pipelines, and agentic AI. Expert in **C#/.NET, TypeScript, Rust, Python, Golang, Java, and Supabase/PostgreSQL**.

26+

Open-Source Repos

7+

Featured Projects

12+

Languages & Runtimes

8

Embedded Mini-Games

FEATURED PROJECT SHOWCASE

RetailHub

↔ github.com/caleheinz25/retailhub

Retail Management & Cross-Platform POS Platform

Tauri

Rust

TypeScript

Supabase

.NET

TanStack Router

- Cross-platform desktop Point of Sale (POS) & inventory management system using Tauri (Rust runtime) + TypeScript frontend.
- Supabase PostgreSQL architecture with Row Level Security (RLS), PL/pgSQL stored functions, and real-time multi-terminal sync.

Developer Portfolio & Interactive Dev Suite

↔ mhisyam.com

Astro 5 Island Architecture with 8 Embedded Mini-Games Arcade

Astro 5

TypeScript

SolidJS

Tailwind CSS 4

HTML5 Canvas

- Ultra-fast static portfolio with embedded Unix terminal shell (`hisyam-cli`), REST API demo client, and audio music player.
- 8 playable mini-games: 2048, Whack-a-Bug, Sudoku, Pong, Minesweeper, Snake, Memory Match, Typing Speed.
- Automated build pipeline fetching live GitHub profile stats and repository telemetry.

Pneumonia U-Net Segmentation Pipeline

↔ [Repository](#)

Medical AI — Deep Learning Semantic Segmentation

Python

PyTorch

U-Net

OpenCV

NumPy

- End-to-end medical image segmentation pipeline using PyTorch U-Net encoder-decoder architecture for chest X-ray analysis.
- Evaluated with Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) metrics.

Gemini Model Context Protocol (MCP) Client

↔ [Repository](#)

Agentic AI Tooling — Structured Workflow Automation

TypeScript

Gemini API

MCP Protocol

Node.js

- Extensible MCP client enabling agentic AI tool-use integrations and structured workflow orchestration with Google Gemini.

Golang & SvelteKit Real-Time Chat

[↔ Repository](#)

Concurrent WebSocket Backend + Reactive SvelteKit Frontend

Golang

Goroutines

WebSockets

SvelteKit

- Multi-room concurrent WebSocket chat server built with Go channels & goroutines, paired with reactive SvelteKit UI.

Secure Java JWT Authentication Microservice

[↔ Repository](#)

Enterprise-Grade Auth — Access + Refresh + Version Tokens

Java

Spring Boot

Spring Security

JWT

PostgreSQL

- Enterprise JWT auth microservice: access tokens, sliding session refresh tokens, and instant token version invalidation.

• TECHNICAL ARSENAL

Languages & Platforms

C#

TypeScript

Rust

Python

Golang

Java

PHP

PL/pgSQL

Backend & APIs

.NET

Go Gin/Fiber

Spring Boot

Node.js

Express.js

Laravel

Desktop & Frontend

Tauri

Astro 5

React.js

SolidJS

SvelteKit

Next.js

Vue.js

Database

Supabase

PostgreSQL

MongoDB

Redis

DevOps & Toolchain

Docker

GitHub Actions

Linux

Bun

CI/CD

AI & Deep Learning

PyTorch

Gemini API

U-Net

OpenCV

MCP Proto

Nginx

• EDUCATION

Universitas Gunadarma

Bekasi, Indonesia

Bachelor of Computer Science (Informatika) — Fresh Graduate

Focus on Software Engineering, Web Systems, and Applied Artificial Intelligence. Maintainer of 26+ open-source repositories spanning AI, web apps, desktop tools, and backend APIs.